Product Requirements Document

Detection and Characterization of Subsurface Ice in Lunar South Polar Regions Using Chandrayaan-2 Radar and Imagery Data for Landing Site and Rover Traverse Planning

1. Product Vision

Build a decision-support platform that fuses Chandrayaan-2 DFSAR radar, OHRC/TMC-2 imagery, terrain, illumination, and scientific priors to identify candidate subsurface ice zones in lunar south-polar permanently shadowed regions and convert them into actionable landing-site and rover-traverse recommendations.

2. Success Definition

- A judge can select a lunar south-pole area and immediately see ice-likelihood, terrain hazards, landing safety, and traverse feasibility.
- The system explains every recommendation using visible evidence layers rather than presenting a black-box score.
- A demo can run on preprocessed public sample data even if live PRADAN downloads are not available during judging.

3. Users

User	Primary Need	Product Response
Mission planner	Shortlist safe and useful landing zones	Ranked sites with safety, science value, and resource-potential scores
Planetary scientist	Inspect evidence for possible subsurface ice	Layer-by-layer radar, morphology, slope, shadow, and confidence views
Rover operations team	Plan traverses that avoid hazards	Cost-aware path planning with slope, illumination, communication, and science targets
Hackathon judge	Understand value quickly	Interactive dashboard, explanation panel, and downloadable mission brief

4. Core Features

Feature	MVP Scope	Stretch Scope
AOI explorer	South-pole dashboard with DFSAR, TMC-2, OHRC context, LOLA validation, cold-trap, and route layers	Draw custom AOI and fetch matching PDS products
Ice likelihood	Evidence-weighted candidate score using radar ratios, cold-trap proxy, and terrain context	Bayesian uncertainty and calibration against published candidate craters
Landing suitability	TMC-2 slope/accessibility with LOLA regional sanity check	Multi-objective optimization for several mission profiles
Rover traverse	Computed A* path from LZ-A to SCI-B over accessibility and cold-trap cost grids	Energy-aware route with communication and thermal constraints
Explainability	Dynamic metrics and processing-chain text update with the selected layer	Counterfactual comparison between candidate sites

Feature	MVP Scope	Stretch Scope
Judge Demo Mode	Guided seven-step story from problem to recommendation	Branching demo paths for science, engineering, and product judges
Reports	PDF/HTML mission brief for selected site	Versioned science audit package

5. Current Dashboard Capability

Capability	Current implementation	Judge value
LOLA validation	Independent NASA PGDA/LRO LOLA slope, roughness, and terrain-class layers are cropped to the TMC-2 AOI.	Shows that the terrain pipeline is being checked against a trusted external reference, not only self-generated layers.
Judge Demo Mode	Seven guided steps walk through the problem, DFSAR evidence, TMC-2 safety, LOLA validation, cold-trap proxy, A* traverse, and final recommendation.	Turns the dashboard into a crisp story that can be presented reliably under time pressure.
A* route	Route from LZ-A to SCI-B is computed from accessibility and cold-trap cost layers instead of being hand drawn.	Connects science target selection to operational rover planning.
Dynamic metrics	Confidence ring, candidate ranking, traverse bars, plan label, and processing-chain text change with the selected layer.	Makes the evidence stack feel alive and auditable instead of static.
Source ledger	Every source is marked as used, validation, context, roadmap, or deprecated with a concrete contribution.	Builds credibility and prevents overclaiming.

6. MVP Workflow

- Load the curated lunar south-pole AOI assembled from official Chandrayaan-2 DFSAR, TMC-2, and OHRC files.
- Normalize and render derived layers: SAR candidate evidence, terrain slope/accessibility, illumination/shadow proxies, cold-trap proxy, LOLA validation, and route focus image.
- Use Judge Demo Mode to step through problem, evidence, validation, traverse, and final site recommendation.
- Compute candidate scores and dynamic metrics that change as the layer stack changes.
- Generate a rover traverse from landing point LZ-A to high-interest science target SCI-B.
- Produce explanation panels, source ledger, and mission-planning PDF report.

7. Scoring Model

The MVP is framed as an evidence-weighted decision system, not a claim of confirmed ice. The current SAR candidate evidence score combines co-pol brightness, cross/co ratio, and HH/VV ratio. Terrain accessibility combines low-slope and local-relief behavior from TMC-2. The cold-trap proxy combines shadow persistence and inverse illumination from low-sun hillshade sweeps. ISRO's 2026 DFSAR release highlights CPR > 1 with DOP < 0.13 as a refined criterion for volumetric scattering potentially associated with subsurface ice; exact CPR/DOP computation remains a validation upgrade.

8. Validation and Scientific Honesty

External validation now uses NASA PGDA/LRO LOLA south-pole slope, roughness, and terrain-class products cropped to the Chandrayaan-2 TMC-2 AOI. The TMC-derived mean slope is 10.67 deg, the LOLA mean slope is 11.63 deg, and the mean difference is 0.96 deg. Because LOLA is 1000 m/pixel, this is a regional sanity check, not meter-scale hazard validation.

The prototype route from LZ-A to SCI-B is generated with A* over downsampled TMC-2 accessibility and cold-trap proxy rasters. The route has 50 path pixels and a relative cost of 151.74. It is a screening route for judge demonstration and needs geodetic calibration, obstacle inflation, and rover dynamics before mission use.

- Use candidate, evidence-consistent, or prioritized target language unless a claim is externally validated.
- Separate raw evidence, derived proxy, validation layer, and operational recommendation in the UI.

9. Source-to-Evidence Ledger

Source	Status	Contribution
ISRO PRADAN	Used	Official Chandrayaan-2 DFSAR, TMC-2, and OHRC product downloads.
Chandrayaan-2 Map Browse	Used	AOI/product discovery and south-pole footprint selection.
DFSAR science release	Science anchor	CPR/DOP interpretation target; current prototype uses ratio-based candidate evidence.
TMC-2 south-pole DTM/orthobrowse	Used	Slope, accessibility, hillshade, cold-trap proxy, and route cost layers.
OHRC	Context	High-resolution visual hazard context until exact footprint registration is complete.
NASA PGDA/LRO LOLA	Used for validation	Independent 1000 m/pixel slope, roughness, and class sanity check.
USGS ISIS	Roadmap	Production-grade planetary projection/calibration path documented for later hardening.

10. Acceptance Criteria

- Map renders in under 5 seconds for preprocessed demo AOIs.
- Each candidate site has a reproducible score breakdown and source-layer thumbnails.
- Judge Demo Mode completes the problem-to-recommendation story without requiring live portal access.
- Dynamic metric panels update when switching between DFSAR, TMC-2, LOLA, cold-trap, and traverse layers.
- Traverse avoids cells above configured slope/hazard thresholds.
- All public data sources and preprocessing assumptions are cited in the UI and report.
- No output uses the phrase confirmed ice unless backed by external labeled truth; use likely, candidate, or evidence-consistent.

11. Risks

Risk	Impact	Mitigation
Limited labeled ground truth	Model validation is difficult	Use physics-informed scoring, published craters, uncertainty, and transparent assumptions
Large PDS products	Downloads and preprocessing may be slow	Curate small AOI bundles and cache COG/Zarr-ready derivatives
Radar false positives from rough terrain	Misleading ice score	Combine DOP, CPR, morphology, slope, and multi-frequency consistency
LOLA resolution mismatch	External validation can be overinterpreted	Clearly call it a regional sanity check, not pixel-level hazard proof
Hackathon time pressure	Overbuilt backend	Ship a strong preprocessed demo first, live ingestion second

Source Links

ISRO Science Data Archive - Chandrayaan-2 PRADAN: https://pradan.issdc.gov.in/ch2/

ISSDC Chandrayaan-2 mission data page: https://www.issdc.gov.in/chandrayaan2.html

ISRO DFSAR subsurface-ice release, May 27, 2026: https://www.isro.gov.in/Chandrayaan2_Dual_Frequency_Synthetic_Aperture_Radar.html

Chandrayaan-2 Map Browse: https://chmapbrowse.issdc.gov.in

VEDAS note on downloading Chandrayaan-2 OHRC products: https://vedas.sac.gov.in/static/pdf/SIH_2024/SIH1732_CH2_PS.pdf

NASA Planetary Data System Geosciences Node: https://pds-geosciences.wustl.edu/

NASA PGDA LRO LOLA south-pole products: https://pgda.gsfc.nasa.gov/products/90

USGS ISIS planetary image processing: https://isis.astrogeology.usgs.gov/